

ENGR 213 · Applied Ordinary Differential Equations

Quiz 1 Practice Exam & Master Solutions Guide (All Topics Up to Exact Equations)

Concordia University · Dept. of Building, Civil & Environmental Engineering · Universal Expansion Standard

ENGR 213: Applied Ordinary Differential Equations

Quiz 1 Master Solutions & Comprehensive Theory Guide

Concordia University · Department of Building, Civil & Environmental Engineering (BCEE)

Coverage: Quiz 1 Preparation (Zill Sections 1.1, 1.2, 2.1, 2.2, 2.3 · Dr. Haghighat Lectures 1–3 · Tutorial 1)

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Part I: Official Tutorial 1 Handout Solutions

Problem 1 · Verification of Solution Family

Problem Statement

(Tutorial 1 Handout · Section 1.1 Exercise 25)

Verify by direct differentiation that the two-parameter family of functions

$$y(x) = c_1 e^{2x} + c_2 x e^{2x}$$

is an explicit solution of the second-order linear differential equation

$$\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 4y = 0$$

on the interval $(-\infty, \infty)$. State the order of the equation and specify whether it is linear or non-linear.

Step-by-Step Solution

Step 1: Classification

- **Order:** The highest derivative present is $\frac{d^2y}{dx^2}$, so this is a **second-order** ODE.
- **Linearity:** The dependent variable y and its derivatives y' and y'' appear only to the first power, have no products with each other (e.g. no yy'), and their coefficients are constants (1, -4, 4). Thus, the equation is **linear**.

Step 2: Compute the First Derivative $y'(x)$

Given:

$$y(x) = c_1e^{2x} + c_2xe^{2x}$$

Differentiate using the chain rule on the first term and the product rule on the second term:

$$\begin{aligned}\frac{d}{dx} [c_1e^{2x}] &= 2c_1e^{2x} \\ \frac{d}{dx} [c_2xe^{2x}] &= c_2 \left(\frac{d}{dx} [x] \cdot e^{2x} + x \cdot \frac{d}{dx} [e^{2x}] \right) = c_2(e^{2x} + 2xe^{2x})\end{aligned}$$

Summing these gives:

$$y'(x) = 2c_1e^{2x} + c_2e^{2x} + 2c_2xe^{2x} = (2c_1 + c_2)e^{2x} + 2c_2xe^{2x}$$

Step 3: Compute the Second Derivative $y''(x)$

Differentiate $y'(x)$:

$$\begin{aligned}y''(x) &= \frac{d}{dx} [(2c_1 + c_2)e^{2x}] + 2c_2 \frac{d}{dx} [xe^{2x}] \\ &= 2(2c_1 + c_2)e^{2x} + 2c_2(e^{2x} + 2xe^{2x}) \\ &= (4c_1 + 2c_2)e^{2x} + 2c_2e^{2x} + 4c_2xe^{2x} \\ y''(x) &= (4c_1 + 4c_2)e^{2x} + 4c_2xe^{2x}\end{aligned}$$

Step 4: Substitute into the Left-Hand Side (LHS) of the ODE

$$\text{LHS} = y'' - 4y' + 4y$$

Substitute y , y' , and y'' :

$$\text{LHS} = [(4c_1 + 4c_2)e^{2x} + 4c_2xe^{2x}] - 4[(2c_1 + c_2)e^{2x} + 2c_2xe^{2x}] + 4[c_1e^{2x} + c_2xe^{2x}]$$

Group like terms in e^{2x} and xe^{2x} :

$$\text{Term in } e^{2x}: (4c_1 + 4c_2) - 4(2c_1 + c_2) + 4c_1 = 4c_1 + 4c_2 - 8c_1 - 4c_2 + 4c_1 = (4 - 8 + 4)c_1 + (4 - 4)c_2 = 0$$

$$\text{Term in } xe^{2x}: 4c_2 - 4(2c_2) + 4c_2 = 4c_2 - 8c_2 + 4c_2 = (4 - 8 + 4)c_2 = 0$$

Therefore:

$$\text{LHS} = 0 \cdot e^{2x} + 0 \cdot xe^{2x} = 0 = \text{RHS}$$

Since this equality holds identically for all real numbers $x \in (-\infty, \infty)$ and for any arbitrary constants c_1, c_2 , the function $y(x) = c_1 e^{2x} + c_2 x e^{2x}$ is an explicit two-parameter family of solutions. ■

💡 Underlying Material & Intuitive Explanation

[!NOTE] Theory & Concepts Tested:

1. **Definition of a Solution:** A function $\phi(x)$ is a solution to an n -th order ODE on an interval I if ϕ possesses at least n continuous derivatives on I and substituting ϕ and its derivatives into the ODE reduces it to an identity ($0 \equiv 0$).
2. **n -Parameter Families:** A linear ODE of order n has a general solution containing exactly n independent arbitrary constants. Here, order = 2, so the family has 2 parameters (c_1, c_2).
3. **Repeated Roots Intuition:** In Chapter 3, you will see that the characteristic equation for $y'' - 4y' + 4y = 0$ is $r^2 - 4r + 4 = (r - 2)^2 = 0$. The single repeated root $r = 2$ gives the first fundamental solution $y_1 = e^{2x}$. To find a second linearly independent solution, reduction of order multiplies by x , yielding $y_2 = x e^{2x}$.

[!WARNING] Concordia Exam Pitfalls & Grade Deduction Traps:

- **The Product Rule Omission:** Students frequently write $\frac{d}{dx}[x e^{2x}] = e^{2x}$ or $2x e^{2x}$. You **must** show both terms: $e^{2x} + 2x e^{2x}$.
- **Lack of Grouping:** Do not just write " $= 0$ " at the bottom of your page without explicitly factoring e^{2x} and $x e^{2x}$. Concordia graders look specifically for the algebraic cancellation showing $(4 - 8 + 4) = 0$.
- **Interval of Definition:** Exponential functions e^{2x} and polynomials x are infinitely differentiable everywhere. Always state the interval explicitly as $I = (-\infty, \infty)$ or $\mathbb{R}$.

Problem 2 · First-Order Non-Linear IVP

Problem Statement

(Tutorial 1 Handout · Section 1.2 Problem 1)

Given that $y(x) = \frac{1}{1+c_1 e^{-x}}$ is a one-parameter family of solutions of the first-order differential equation

$$y' = y - y^2$$

(a) Find a solution of the first-order IVP consisting of this differential equation and the initial condition:

$$y(0) = -\frac{1}{3}$$

(b) Determine the domain and the exact, maximal interval of definition I for this particular solution.

Step-by-Step Solution

Step 1: Apply the Initial Condition to Solve for c_1

We are given $y(0) = -\frac{1}{3}$. Substitute $x = 0$ and $y = -\frac{1}{3}$ into the given family of solutions:

$$-\frac{1}{3} = \frac{1}{1 + c_1 e^{-(0)}}$$

Since $e^0 = 1$:

$$-\frac{1}{3} = \frac{1}{1 + c_1}$$

Cross-multiply:

$$1 + c_1 = -3 \implies c_1 = -3 - 1 = -4$$

Step 2: Write the Particular Solution

Substitute $c_1 = -4$ back into the solution family:

$$y(x) = \frac{1}{1 - 4e^{-x}}$$

Step 3: Determine the Interval of Definition I

For a function to be a valid solution of an initial value problem, it must be continuous and differentiable on an open interval I containing the initial point $x_0 = 0$. The function $y(x)$ becomes undefined when its denominator equals zero:

$$1 - 4e^{-x} = 0 \implies 4e^{-x} = 1 \implies e^{-x} = \frac{1}{4} \implies e^x = 4$$

Taking the natural logarithm of both sides:

$$x = \ln(4) = 2 \ln(2) \approx 1.3863$$

Thus, the domain of the function is all real numbers except $x = \ln 4$:

$$\text{Domain} = (-\infty, \ln 4) \cup (\ln 4, \infty)$$

However, by mathematical definition, the **interval of definition I of an IVP must be a single connected interval** containing the initial value $x_0 = 0$. Compare $x_0 = 0$ with the singularity $x = \ln 4$:

$$0 < \ln(4)$$

Therefore, the initial point $x = 0$ lies strictly inside the left interval:

$$I = (-\infty, \ln 4)$$

Underlying Material & Intuitive Explanation

[!NOTE] Theory & Concepts Tested:

- Initial Value Problem (IVP):** Finding a specific member of a family of curves that passes through a given coordinate point (x_0, y_0) .
- Interval of Definition (I):** An ODE solution is NOT merely a formula; it is a pair $(\phi(x), I)$ where I is a connected open interval on which $\phi(x)$ satisfies the ODE. A disconnected union of sets is never an interval of definition for an IVP.
- The Logistic / Bernoulli Structure:** The equation $y' = y - y^2$ is the standard autonomous logistic equation with $r = 1, K = 1$. When $y(0) = -1/3 < 0$, the population is negative (outside the physical carrying capacity), so the solution blows up to $-\infty$ in finite time at $x = \ln 4$.

[!WARNING] Concordia Exam Pitfalls & Grade Deduction Traps:

- The Union Set Trap:** Writing $x \neq \ln 4$ or $(-\infty, \ln 4) \cup (\ln 4, \infty)$ as the answer to "interval of definition" will cost you 2 out of 5 marks! An interval of definition must be connected and contain x_0 .
- Sign Error in Exponents:** Don't confuse $e^{-x} = 1/4$ with $e^x = 1/4$. Double-check: $e^{-x} = 1/4 \implies -x = \ln(1/4) = -\ln 4 \implies x = \ln 4$.

Problem 3 · Second-Order Linear IVP

Problem Statement

(Tutorial 1 Handout · Section 1.2 Problem 11)

Given that $y(x) = c_1e^x + c_2e^{-x}$ is a two-parameter family of solutions of the second-order differential equation

$$y'' - y = 0$$

Find a solution of the second-order IVP consisting of this differential equation and the given initial conditions:

$$y(0) = 1, \quad y'(0) = 2$$

Step-by-Step Solution

Step 1: Differentiate the General Solution

The general solution is:

$$y(x) = c_1e^x + c_2e^{-x}$$

Differentiate with respect to x :

$$y'(x) = c_1e^x - c_2e^{-x}$$

Step 2: Set Up the System of Linear Equations Using the Initial Conditions

Apply the first initial condition $y(0) = 1$:

$$y(0) = c_1e^0 + c_2e^0 = c_1 + c_2 = 1 \quad \text{— (Equation 1)}$$

Apply the second initial condition $y'(0) = 2$:

$$y'(0) = c_1e^0 - c_2e^0 = c_1 - c_2 = 2 \quad \text{— (Equation 2)}$$

Step 3: Solve the Linear System for c_1 and c_2

Add Equation (1) and Equation (2):

$$(c_1 + c_2) + (c_1 - c_2) = 1 + 2$$

$$2c_1 = 3 \implies c_1 = \frac{3}{2}$$

Subtract Equation (2) from Equation (1):

$$(c_1 + c_2) - (c_1 - c_2) = 1 - 2$$

$$2c_2 = -1 \implies c_2 = -\frac{1}{2}$$

Step 4: Formulate the Unique Particular Solution

Substitute $c_1 = \frac{3}{2}$ and $c_2 = -\frac{1}{2}$ into $y(x)$:

$$y(x) = \frac{3}{2}e^x - \frac{1}{2}e^{-x}$$

(Alternative hyperbolic form: $y(x) = \cosh(x) + 2 \sinh(x)$)

Step 5: State the Interval of Definition

Since e^x and e^{-x} are continuous and infinitely differentiable on all of $\mathbb{R}$:

$$I = (-\infty, \infty)$$

Underlying Material & Intuitive Explanation**[!NOTE] Theory & Concepts Tested:**

1. **Second-Order IVP Requirements:** An n -th order linear ODE requires n initial conditions specified at the **exact same point** x_0 (e.g. $y(x_0) = y_0, y'(x_0) = y_1$) to uniquely determine the n constants. If conditions were given at two different points (e.g. $y(0) = 1, y(1) = 2$), it would be a **Boundary Value Problem (BVP)**, which does not guarantee a unique solution.
2. **Superposition Principle:** For linear homogeneous equations, any linear combination of solutions $c_1 y_1 + c_2 y_2$ is also a solution.

[!WARNING] Concordia Exam Pitfalls & Grade Deduction Traps:

- **Forgetting the Negative Sign in y' :** $\frac{d}{dx}[e^{-x}] = -e^{-x}$. Missing this turns $c_1 - c_2 = 2$ into $c_1 + c_2 = 2$, which contradicts Equation 1 and makes the system inconsistent!
- **Checking Your Work:** Always plug $x = 0$ back into your final equation: $y(0) = \frac{3}{2}(1) - \frac{1}{2}(1) = 1$ (Correct!)
 $y'(0) = \frac{3}{2}(1) + \frac{1}{2}(1) = 2$ (Correct!)

Problem 4 · Autonomous Phase Portrait & Stability**Problem Statement**

(Tutorial 1 Handout · Section 2.1 Problem 21)

Consider the autonomous first-order differential equation:

$$\frac{dy}{dx} = y^2 - 3y$$

- (a) Find all critical points (equilibrium solutions).
- (b) Draw the one-dimensional phase line (phase portrait) showing the sign of $\frac{dy}{dx}$ and directional arrows.
- (c) Classify each critical point as asymptotically stable, unstable, or semi-stable.
- (d) By hand, sketch typical solution curves in the regions in the xy -plane determined by the graphs of the equilibrium solutions.

Step-by-Step Solution**Step 1: Find Critical Points**

Set the autonomous rate function $f(y) = 0$:

$$\begin{aligned} f(y) &= y^2 - 3y = y(y - 3) = 0 \\ \implies y &= 0 \quad \text{and} \quad y = 3 \end{aligned}$$

The horizontal equilibrium solutions are the constant functions:

$$y(x) \equiv 0 \quad \text{and} \quad y(x) \equiv 3$$

Step 2: Sign Analysis of $f(y)$ on the Sub-Intervals

The critical points divide the y -axis into three distinct intervals:

1. **Interval $(3, \infty)$:**

- Pick test value $y = 4$: $f(4) = (4)^2 - 3(4) = 16 - 12 = +4 > 0$.
- Since $y' > 0$, solutions are **increasing**. On the phase line, the arrow points **UP** ($\uparrow$).

2. **Interval $(0, 3)$:**

- Pick test value $y = 1$: $f(1) = (1)^2 - 3(1) = 1 - 3 = -2 < 0$.
- Since $y' < 0$, solutions are **decreasing**. On the phase line, the arrow points **DOWN** ($\downarrow$).

3. **Interval $(-\infty, 0)$:**

- Pick test value $y = -1$: $f(-1) = (-1)^2 - 3(-1) = 1 + 3 = +4 > 0$.
- Since $y' > 0$, solutions are **increasing**. On the phase line, the arrow points **UP** ($\uparrow$).

Step 3: Construct the 1D Phase Line & Classify Stability

y-axis	Sign of y'	Arrow	Classification
$y > 3$	$f(y) > 0$	$\uparrow$	UNSTABLE (Repeller / Source)
$y = 3$		$\bullet$	
$0 < y < 3$	$f(y) < 0$	$\downarrow$	ASYMPTOTICALLY STABLE (Attractor / Sink)
$y = 0$		$\bullet$	
$y < 0$	$f(y) > 0$	$\uparrow$	

- **At $y = 3$:** Solutions for $y > 3$ move upward away from 3 ($\uparrow$); solutions for $0 < y < 3$ move downward away from 3 ($\downarrow$). Because trajectories diverge away from $y = 3$ on both sides, $y = 3$ is **UNSTABLE (Repeller / Source)**.
- **At $y = 0$:** Solutions for $0 < y < 3$ move downward toward 0 ($\downarrow$); solutions for $y < 0$ move upward toward 0 ($\uparrow$). Because trajectories converge toward $y = 0$ from both sides, $y = 0$ is **ASYMPTOTICALLY STABLE (Attractor / Sink)**.

Step 4: Concavity Analysis & Solution Curve Sketching

To sketch realistic curves, find y'' using the chain rule:

$$y'' = \frac{d}{dx}[f(y)] = f'(y) \frac{dy}{dx} = f'(y)f(y)$$

$$f'(y) = \frac{d}{dy}[y^2 - 3y] = 2y - 3$$

$$y'' = (2y - 3)(y^2 - 3y) = (2y - 3)y(y - 3)$$

Setting $y'' = 0$ gives an **inflection point** at $y = \frac{3}{2} = 1.5$.

- In $(3, \infty)$: $(+) \cdot (+) \cdot (+) > 0 \implies y'' > 0$ (**Concave Up**). Solutions increase rapidly toward $+\infty$.
- In $(1.5, 3)$: $(+) \cdot (+) \cdot (-) < 0 \implies y'' < 0$ (**Concave Down**).
- In $(0, 1.5)$: $(-) \cdot (+) \cdot (-) > 0 \implies y'' > 0$ (**Concave Up**).
- In $(-\infty, 0)$: $(-) \cdot (-) \cdot (-) < 0 \implies y'' < 0$ (**Concave Down**). Solutions increase asymptotically toward the line $y = 0$ as $x \rightarrow \infty$.

💡 Underlying Material & Intuitive Explanation

[!NOTE] Theory & Concepts Tested:

- Autonomous Differential Equations:** Equations of the form $\frac{dy}{dx} = f(y)$ where the independent variable x does not appear explicitly. This means slopes are identical along any horizontal line in the xy -plane!
- Derivative Test Shortcut for Stability:**
 - If $f'(c) < 0 \implies$ Critical point c is **Asymptotically Stable** (Attractor).
 - If $f'(c) > 0 \implies$ Critical point c is **Unstable** (Repeller).
 - Quick check: $f'(y) = 2y - 3 \implies f'(0) = -3 < 0$ (Stable!), $f'(3) = 2(3) - 3 = +3 > 0$ (Unstable!). This 5-second check guarantees full marks.
- The No-Crossing Rule:** Because $f(y)$ and $f'(y)$ are smooth polynomials, Picard's Uniqueness Theorem holds everywhere. Therefore, **solution curves can never intersect each other, nor can they cross the equilibrium lines $y = 0$ or $y = 3$.**

[!WARNING] Concordia Exam Pitfalls & Grade Deduction Traps:

- Arrow Direction Confusion:** Remember that on a phase line, an arrow points UP ($\uparrow$) when $dy/dx > 0$ (meaning y is increasing), and DOWN ($\downarrow$) when $dy/dx < 0$ (meaning y is decreasing). Never draw arrows horizontally!
- Neglecting Equilibrium Solutions:** When asked for the sketch, you must explicitly draw the horizontal dashed or solid equilibrium lines $y = 0$ and $y = 3$ first. Graders deduct marks if these asymptotes are missing.

Problem 5 · Autonomous Factorization & Concavity

Problem Statement

(Tutorial 1 Handout · Section 2.1 Problem 24)

Consider the autonomous first-order differential equation:

$$\frac{dy}{dx} = 10 + 3y - y^2$$

- Find all critical points.
- Construct the 1D phase portrait and classify each critical point as asymptotically stable, unstable, or semi-stable.
- Using $y'' = f'(y)f(y)$, determine the inflection point and the concavity behavior of the solution curves.
- If an initial condition is given as $y(0) = 1$, determine the asymptotic limit $\lim_{x \rightarrow \infty} y(x)$ without solving the ODE.

Step-by-Step Solution

Step 1: Factor $f(y)$ to Find Critical Points

Set $f(y) = 0$:

$$\begin{aligned} 10 + 3y - y^2 = 0 &\implies -(y^2 - 3y - 10) = 0 \\ -(y - 5)(y + 2) &= (5 - y)(y + 2) = 0 \\ \implies y = 5 &\text{ and } y = -2 \end{aligned}$$

The equilibrium solutions are $y(x) \equiv 5$ and $y(x) \equiv -2$.

Step 2: Sign Analysis of $f(y)$

- Region $y > 5$:**

◦ Test $y = 6$: $f(6) = (5 - 6)(6 + 2) = (-1)(8) = -8 < 0 \implies$ Arrow points **DOWN** ($\downarrow$).

2. **Region** $-2 < y < 5$:

◦ Test $y = 0$: $f(0) = 10 + 0 - 0 = +10 > 0 \implies$ Arrow points **UP** ($\uparrow$).

3. **Region** $y < -2$:

◦ Test $y = -3$: $f(-3) = (5 - (-3))(-3 + 2) = (8)(-1) = -8 < 0 \implies$ Arrow points **DOWN** ($\downarrow$).

Step 3: Phase Line & Classification

- **At** $y = 5$: Trajectories from below point UP ($\uparrow$), and trajectories from above point DOWN ($\downarrow$). Trajectories converge toward $y = 5$ from both directions.

$y = 5$ is **ASYMPTOTICALLY STABLE** (Attractor / Sink)

- **At** $y = -2$: Trajectories for $-2 < y < 5$ point UP ($\uparrow$), and trajectories for $y < -2$ point DOWN ($\downarrow$). Trajectories diverge away from $y = -2$ on both sides.

$y = -2$ is **UNSTABLE** (Repeller / Source)

(Derivative test verification: $f'(y) = 3 - 2y \implies f'(5) = 3 - 10 = -7 < 0$ (Stable); $f'(-2) = 3 - 2(-2) = +7 > 0$ (Unstable).)

Step 4: Inflection Point & Concavity

$$y'' = f'(y)f(y) = (3 - 2y)(10 + 3y - y^2) = -(2y - 3)(5 - y)(y + 2)$$

Setting $f'(y) = 0$:

$$3 - 2y = 0 \implies y = \frac{3}{2} = 1.5$$

The inflection point in the middle region occurs at $y = 1.5$.

- For $1.5 < y < 5$: $f'(y) < 0$ and $f(y) > 0 \implies y'' < 0$ (**Concave Down**).
- For $-2 < y < 1.5$: $f'(y) > 0$ and $f(y) > 0 \implies y'' > 0$ (**Concave Up**).

Step 5: Asymptotic Limit for $y(0) = 1$

The initial value $y(0) = 1$ falls strictly inside the interval $(-2, 5)$. In this interval, $f(y) > 0$, so $y(x)$ is strictly increasing. Since $y = 5$ is an asymptotically stable attractor and solution curves cannot cross equilibrium lines:

$$\lim_{x \rightarrow \infty} y(x) = 5$$

$$\lim_{x \rightarrow -\infty} y(x) = -2$$

Underlying Material & Intuitive Explanation

[!NOTE] Theory & Concepts Tested:

1. **Quadratic Negative Sign Trap:** A common mistake is factoring $10 + 3y - y^2$ as $(y - 5)(y + 2)$. The coefficient of y^2 is -1 , so the factors are $-(y - 5)(y + 2)$. Forgetting the minus sign completely reverses the stability classification!
2. **S-Shaped (Sigmoid) Curves:** Between the unstable source $y = -2$ and the stable sink $y = 5$, solutions exhibit a characteristic logistic "S" shape: they accelerate upward while concave up ($y < 1.5$), pass through the inflection point at $y = 1.5$ where growth rate is maximized, and then decelerate while concave down ($y > 1.5$) as they approach the carrying capacity $y = 5$.

Part II: Teacher Lecture Slide Core Exam Solutions

Problem 6 · Picard's Existence & Uniqueness Theorem

Problem Statement

(Dr. Haghighat Lecture 2 · Slide 8)

Consider the first-order initial value problem:

$$\frac{dy}{dx} = x\sqrt{y}, \quad y(x_0) = y_0$$

- (a) State the conditions of Picard's Existence and Uniqueness Theorem for a first-order IVP.
- (b) Explain why the theorem **fails** to guarantee uniqueness for the initial condition $y(0) = 0$. Show by direct substitution that both $y_1(x) \equiv 0$ and $y_2(x) = \frac{1}{16}x^4$ ($x \geq 0$) are valid solutions passing through $(0, 0)$.
- (c) Does Picard's theorem guarantee a unique solution for the initial condition $y(2) = 1$? Justify your answer mathematically.

Step-by-Step Solution

Step 1: Formal Statement of Picard's Theorem

Let R be a rectangular region in the xy -plane defined by $a \leq x \leq b$; $c \leq y \leq d$ that contains the point (x_0, y_0) in its interior. If:

1. $f(x, y)$ is **continuous** on R , and
2. $\frac{\partial f}{\partial y}$ is **continuous** on R , then there exists an open interval $I_0 = (x_0 - h, x_0 + h)$ with $h > 0$ contained in $[a, b]$, on which there exists a **unique** function $y(x)$ that satisfies the initial value problem $\frac{dy}{dx} = f(x, y)$; $y(x_0) = y_0$.

Step 2: Evaluation of the Functions for the Given ODE

Here:

$$f(x, y) = xy^{1/2}$$

- Continuity of f : $f(x, y)$ is continuous on the upper half-plane:

$$(x, y) \in \mathbb{R}^2 \mid y \geq 0$$

Now compute the partial derivative with respect to y :

$$\frac{\partial f}{\partial y} = \frac{\partial}{\partial y} [xy^{1/2}] = x \cdot \frac{1}{2}y^{-1/2} = \frac{x}{2\sqrt{y}}$$

- Continuity of $\frac{\partial f}{\partial y}$: The term $\frac{x}{2\sqrt{y}}$ requires $y > 0$ to avoid division by zero. Thus, $\frac{\partial f}{\partial y}$ is continuous **only** on the open region:

$$(x, y) \in \mathbb{R}^2 \mid y > 0$$

Along the line $y = 0$ (the x -axis), $\frac{\partial f}{\partial y}$ is undefined and therefore discontinuous.

Step 3: Analysis for Initial Condition $y(0) = 0$

The point $(x_0, y_0) = (0, 0)$ lies directly on the boundary $y = 0$. Any open rectangle R centered at $(0, 0)$ must contain points where $y \leq 0$, which includes $y = 0$ where $\frac{\partial f}{\partial y}$ does not exist. Since condition (2) of Picard's theorem is violated, **the theorem guarantees existence (since f is continuous), but fails to guarantee uniqueness.**

Verification of Multiple Solutions Through $(0, 0)$:

- **Candidate 1:** $y_1(x) \equiv 0$.
 - $y_1(0) = 0$.
 - $\frac{dy_1}{dx} = 0$.
 - $x\sqrt{y_1} = x\sqrt{0} = 0$.
 - $0 = 0 \implies$ Valid solution!
- **Candidate 2:** $y_2(x) = \frac{1}{16}x^4$ ($x \geq 0$).
 - $y_2(0) = \frac{1}{16}(0)^4 = 0$.
 - $\frac{dy_2}{dx} = \frac{4}{16}x^3 = \frac{1}{4}x^3$.
 - $x\sqrt{y_2} = x\sqrt{\frac{1}{16}x^4} = x \cdot \left(\frac{1}{4}x^2\right) = \frac{1}{4}x^3$.
 - $\frac{dy_2}{dx} = x\sqrt{y_2} \implies \frac{1}{4}x^3 = \frac{1}{4}x^3 \implies$ Valid solution! Since $y_1(x) \neq y_2(x)$ for $x > 0$, **uniqueness is broken at $(0, 0)$.**

Step 4: Analysis for Initial Condition $y(2) = 1$

Here $(x_0, y_0) = (2, 1)$. The point $(2, 1)$ has $y_0 = 1 > 0$. We can easily construct a rectangle R around $(2, 1)$ that stays strictly above the x -axis, for instance:

$$R = (x, y) \mid 1 < x < 3; 0.5 < y < 1.5$$

On this rectangle:

1. $f(x, y) = x\sqrt{y}$ is a composition of continuous functions, hence **continuous**.
2. $\frac{\partial f}{\partial y} = \frac{x}{2\sqrt{y}}$ has no zero denominators because $y \geq 0.5 > 0$, hence **continuous**. Because both conditions are satisfied on R , Picard's Theorem **guarantees that there exists a unique solution** to the IVP on some open interval containing $x_0 = 2$.

💡 Underlying Material & Intuitive Explanation

[!NOTE] Theory & Concepts Tested:

1. Existence vs. Uniqueness:

- Continuity of $f(x, y)$ alone guarantees that at least ONE solution exists (Peano's Theorem).
- Continuity of BOTH $f(x, y)$ and $\frac{\partial f}{\partial y}$ guarantees that PRECISELY ONE (unique) solution exists.

2. Physical Meaning of Uniqueness Breaking:

If uniqueness breaks, solution curves can branch out or touch each other. Here, the non-trivial curve $y = \frac{1}{16}x^4$ tangentially touches the horizontal solution $y \equiv 0$ at $x = 0$.

[!WARNING] **Concordia Exam Pitfalls & Grade Deduction Traps:**

- **The "Theorem Proves No Solution" Error:** If $\frac{\partial f}{\partial y}$ is discontinuous, Picard's theorem is **inconclusive**; it does NOT say "a unique solution does not exist." It only says uniqueness is not *guaranteed*. You must state that explicitly to get full marks.
- **Check Points Carefully:** When given (x_0, y_0) , always look immediately at y_0 . If y_0 makes any denominator in $\frac{\partial f}{\partial y}$ zero, uniqueness is not guaranteed!

Problem 7 · Second-Order Harmonic Oscillator IVP**Problem Statement**

(Dr. Haghighat Lecture 2 · Slide 5)

Given that $x(t) = c_1 \cos(4t) + c_2 \sin(4t)$ is the general solution of the second-order differential equation

$$x'' + 16x = 0$$

Solve the initial value problem with the initial conditions:

$$x\left(\frac{\pi}{2}\right) = -2, \quad x'\left(\frac{\pi}{2}\right) = 1$$

Step-by-Step Solution**Step 1: Differentiate $x(t)$**

Given:

$$x(t) = c_1 \cos(4t) + c_2 \sin(4t)$$

Using the chain rule $\left(\frac{d}{dt}[4t] = 4\right)$:

$$x'(t) = -4c_1 \sin(4t) + 4c_2 \cos(4t)$$

Step 2: Apply the First Initial Condition $x(\pi/2) = -2$ Substitute $t = \frac{\pi}{2}$:

$$4t = 4\left(\frac{\pi}{2}\right) = 2\pi$$

Recall the trigonometric values at 2π :

$$\cos(2\pi) = 1, \quad \sin(2\pi) = 0$$

Substitute these into the equation for $x(t)$:

$$-2 = c_1 \cos(2\pi) + c_2 \sin(2\pi)$$

$$-2 = c_1(1) + c_2(0) \implies c_1 = -2$$

Step 3: Apply the Second Initial Condition $x'(\pi/2) = 1$ Substitute $t = \frac{\pi}{2}$ into $x'(t)$:

$$1 = -4c_1 \sin(2\pi) + 4c_2 \cos(2\pi)$$

$$1 = -4c_1(0) + 4c_2(1)$$

$$1 = 4c_2 \implies c_2 = \frac{1}{4}$$

Step 4: Write the Final Particular Solution

$$x(t) = -2 \cos(4t) + \frac{1}{4} \sin(4t)$$

💡 Underlying Material & Intuitive Explanation

[!NOTE] Theory & Concepts Tested:

- Harmonic Motion:** The ODE $x'' + \omega^2 x = 0$ represents undamped simple harmonic motion with angular frequency $\omega = \sqrt{16} = 4$ rad/s and period $T = \frac{2\pi}{\omega} = \frac{\pi}{2}$ s.
- Evaluation at Periodic Intervals:** Notice that $t = \pi/2$ corresponds to exactly one full period T . At $t = T$, the cosine term returns to its peak (1) and sine is at its zero crossing (0), which decouples the system of equations and allows c_1 and c_2 to be solved instantly!

Problem 8 · Autonomous Logistic Growth

Problem Statement

(Dr. Haghighat Lecture 1 Slide 6 & Lecture 2 Slide 15)

The population dynamics of a species in a constrained ecosystem is modeled by the autonomous differential equation:

$$\frac{dP}{dt} = rP \left(1 - \frac{P}{K} \right)$$

where $r > 0$ is the intrinsic growth rate and $K > 0$ is the carrying capacity.

- Identify the critical points.
- Construct the phase portrait and classify the stability of each equilibrium point.
- Describe the asymptotic behavior of $P(t)$ as $t \rightarrow \infty$ for:

- An initial population $0 < P(0) < K$.
- An initial population $P(0) > K$.

Step-by-Step Solution

Step 1: Critical Points

Set $f(P) = 0$:

$$rP \left(1 - \frac{P}{K} \right) = 0$$

Since $r > 0$, this yields:

$$P = 0 \quad \text{and} \quad P = K$$

Step 2: Sign of $\frac{dP}{dt}$ in Each Region

- $P > K$: $P > 0$, but $\left(1 - \frac{P}{K}\right) < 0 \implies \frac{dP}{dt} < 0$. Trajectories point **DOWN** (↓).
- $0 < P < K$: $P > 0$ and $\left(1 - \frac{P}{K}\right) > 0 \implies \frac{dP}{dt} > 0$. Trajectories point **UP** (↑).

3. $P < 0$: (Unphysical for population, but mathematically): $P < 0$ and $\left(1 - \frac{P}{K}\right) > 0 \implies \frac{dP}{dt} < 0$. Trajectories point **DOWN** ($\downarrow$).

Step 3: Phase Line & Stability Classification

- At $P = 0$: Trajectories move away from 0 (downward for $P < 0$, upward for $P > 0$).

$P = 0$ is UNSTABLE (Repeller)

- At $P = K$: Trajectories move upward toward K from below, and downward toward K from above.

$P = K$ is ASYMPTOTICALLY STABLE (Attractor)

Step 4: Long-Term Behavior Analysis

1. If $0 < P(0) < K$:

- Since $\frac{dP}{dt} > 0$, the population monotonically increases.
- As $t \rightarrow \infty$, the trajectory approaches the stable equilibrium:

$$\lim_{t \rightarrow \infty} P(t) = K$$

- The growth accelerates until the inflection point $P = K/2$, after which resource limitations cause growth to decelerate.

2. If $P(0) > K$:

- Since $\frac{dP}{dt} < 0$, the population exceeds the carrying capacity (overcrowding, food scarcity) and monotonically decreases.
- As $t \rightarrow \infty$:

$$\lim_{t \rightarrow \infty} P(t) = K$$

💡 Underlying Material & Intuitive Explanation

[!NOTE] **Theory & Concepts Tested:**

1. **Carrying Capacity (K)**: The maximum sustainable population supported by the environment.
2. **Inflection Point Derivation:**

$$\frac{d^2P}{dt^2} = \frac{d}{dP} \left[rP - \frac{r}{K}P^2 \right] \frac{dP}{dt} = \left(r - \frac{2r}{K}P \right) \frac{dP}{dt} = r \left(1 - \frac{2P}{K} \right) \frac{dP}{dt}$$

Setting $\frac{d^2P}{dt^2} = 0$ for non-equilibrium solutions gives $1 - \frac{2P}{K} = 0 \implies P = \frac{K}{2}$. The maximum population growth rate occurs exactly at half of the carrying capacity.

Problem 9 · Separable DE & Lost Singular Solutions

Problem Statement

(Dr. Haghighat Lecture 3 · Slide 6)

Consider the first-order differential equation:

$$\frac{dy}{dx} = y^2 - 4$$

- (a) Identify all constant equilibrium solutions before separating variables.
- (b) Using separation of variables and partial fraction decomposition, obtain the one-parameter family of solutions.

(c) Determine whether any equilibrium solution is a **singular (lost) solution** that cannot be obtained from the general family for any finite real constant c .

Step-by-Step Solution

Step 1: Find Constant Solutions

Before dividing by $(y^2 - 4)$, find where it vanishes:

$$y^2 - 4 = 0 \implies (y - 2)(y + 2) = 0 \implies y = 2 \quad \text{and} \quad y = -2$$

Both $y(x) \equiv 2$ and $y(x) \equiv -2$ are valid constant solutions.

Step 2: Separate Variables (for $y \neq \pm 2$)

$$\frac{dy}{y^2 - 4} = dx$$

Step 3: Partial Fraction Decomposition

$$\frac{1}{y^2 - 4} = \frac{1}{(y - 2)(y + 2)} = \frac{A}{y - 2} + \frac{B}{y + 2}$$

$$1 = A(y + 2) + B(y - 2)$$

- Setting $y = 2$: $1 = A(4) \implies A = \frac{1}{4}$
- Setting $y = -2$: $1 = B(-4) \implies B = -\frac{1}{4}$ Thus:

$$\frac{1}{4} \left(\frac{1}{y - 2} - \frac{1}{y + 2} \right) dy = dx$$

Step 4: Integrate Both Sides

$$\frac{1}{4} \int \left(\frac{1}{y - 2} - \frac{1}{y + 2} \right) dy = \int dx$$

$$\frac{1}{4} (\ln |y - 2| - \ln |y + 2|) = x + C_1$$

Multiply by 4 and combine logarithms:

$$\ln \left| \frac{y - 2}{y + 2} \right| = 4x + 4C_1$$

Exponentiate:

$$\left| \frac{y - 2}{y + 2} \right| = e^{4C_1} e^{4x}$$

Remove absolute values by defining $c = \pm e^{4C_1} \neq 0$:

$$\frac{y - 2}{y + 2} = ce^{4x}$$

Step 5: Solve Explicitly for $y(x)$

$$y - 2 = ce^{4x}(y + 2) = ce^{4x}y + 2ce^{4x}$$

$$y - ce^{4x}y = 2 + 2ce^{4x}$$

$$y(1 - ce^{4x}) = 2(1 + ce^{4x})$$

$$y(x) = \frac{2(1 + ce^{4x})}{1 - ce^{4x}}$$

Step 6: Test Equilibrium Solutions for Singular Status

- **Test $y = 2$:** If we set $c = 0$ in the formula:

$$y(x) = \frac{2(1 + 0)}{1 - 0} = 2$$

Since setting the arbitrary constant $c = 0$ reproduces $y = 2$, this solution is **included in the family**.

- **Test $y = -2$:** Can $y(x) = -2$ for some real number c ?

$$\frac{2(1 + ce^{4x})}{1 - ce^{4x}} = -2$$

$$2(1 + ce^{4x}) = -2(1 - ce^{4x})$$

$$1 + ce^{4x} = -1 + ce^{4x}$$

Subtracting ce^{4x} from both sides yields:

$$1 = -1 \quad (\text{Contradiction!})$$

No finite real constant c can ever generate the solution $y(x) \equiv -2$ (it corresponds to the formal limit $c \rightarrow \infty$).

$y(x) \equiv -2$ is a SINGULAR (LOST) SOLUTION

💡 Underlying Material & Intuitive Explanation

[!NOTE] Theory & Concepts Tested:

1. **Singular Solution Definition:** A solution to an ODE that cannot be obtained from the general n -parameter family by assigning any finite real values to the constants.
2. **Where Lost Solutions Come From:** In step 2, dividing by $(y^2 - 4)$ is mathematically invalid when $y = \pm 2$. Thus, those solutions are stripped away from the algebra and must be tested manually at the end.

[!WARNING] Concordia Exam Pitfalls & Grade Deduction Traps:

- **The Partial Fractions Omission:** Don't guess the integral of $\frac{1}{y^2 - 4}$. Writing $\arctan$ or $\ln(y^2 - 4)$ is a severe error. You must use partial fractions: $\frac{1}{4} \ln \left| \frac{y-2}{y+2} \right|$.
- **Absolute Values:** Always maintain absolute values $\ln | \dots |$ until you absorb the $\pm$ into the constant $c = \pm e^{4C_1}$.

Problem 10 · Separable IVP & Explicit Branch Selection

Problem Statement

(Dr. Haghighat Lecture 3 · Slide 5)

Solve the initial value problem:

$$\frac{dy}{dx} = -\frac{x}{y}, \quad y(4) = -3$$

- (a) Find the explicit particular solution $y(x)$, explaining clearly how the initial condition determines the choice of sign ($\pm$).
- (b) State the geometric curve represented by the implicit family.
- (c) Determine the maximal open interval of definition I on which this particular solution is valid and differentiable.
-

Step-by-Step Solution

Step 1: Separate Variables & Integrate

$$y, dy = -x, dx$$

$$\int y, dy = - \int x, dx$$

$$\frac{1}{2}y^2 = -\frac{1}{2}x^2 + C_1$$

Multiply by 2:

$$x^2 + y^2 = 2C_1 = C$$

Step 2: Determine C Using the Initial Condition

Substitute $x_0 = 4$ and $y_0 = -3$:

$$(4)^2 + (-3)^2 = C$$

$$16 + 9 = C \implies C = 25$$

The implicit relation is the circle:

$$x^2 + y^2 = 25$$

Step 3: Solve Explicitly for $y(x)$ and Select the Sign Branch

$$y^2 = 25 - x^2 \implies y(x) = \pm\sqrt{25 - x^2}$$

Since the initial condition specifies $y(4) = -3$, which is strictly **negative**, we **must choose the negative branch**:

$$y(x) = -\sqrt{25 - x^2}$$

(Verification: $y(4) = -\sqrt{25 - 16} = -\sqrt{9} = -3$. If you chose $+$, you would get $+3 \neq -3$.)

Step 4: Determine the Maximal Interval of Definition I

For $y(x) = -\sqrt{25 - x^2}$ to be real, we require $25 - x^2 \geq 0 \implies -5 \leq x \leq 5$. However, in the original differential equation:

$$\frac{dy}{dx} = -\frac{x}{y}$$

The dependent variable y appears in the denominator! At $x = \pm 5$, $y = 0$, which causes division by zero. Geometrically, the circle has **vertical tangents** ($\frac{dy}{dx} \rightarrow \pm\infty$) at $x = -5$ and $x = 5$. By definition, a solution to a differential equation must be differentiable on its interval of definition. Therefore, the boundary points must be excluded, giving the **open interval**:

$$I = (-5, 5)$$

 Underlying Material & Intuitive Explanation

[!NOTE] **Theory & Concepts Tested:**

1. **Explicit vs. Implicit Solutions:** An implicit relation like $x^2 + y^2 = 25$ defines two distinct functions: the upper hemisphere $y = +\sqrt{25 - x^2}$ and the lower hemisphere $y = -\sqrt{25 - x^2}$. The initial condition uniquely selects which hemisphere represents the physical trajectory.
2. **Differentiability Requirement:** The definition of an ODE solution requires $\frac{dy}{dx}$ to exist at every point in I . Thus, endpoints with vertical slopes can never be included in I .

Part III: High-Probability Quiz & Homework Essentials

Problem 11 · Complete DE Classification Matrix

Problem Statement

(Zill Section 1.1 Assigned Problems)
Classify each differential equation by Type, Order, Linearity, and identify the dependent and independent variables.

Solution Table

Equation	Type	Order	Linearity	Dependent Variable	Independent Variable	Non-Linear Reason (if applicable)
(a) $(1 - x)y'' - 4xy' + 5y = \cos x$	ODE	2	Linear	y	x	N/A (coefficients depend only on x)
(b) $\frac{d^3y}{dx^3} + y\frac{dy}{dx} = e^x$	ODE	3	Non-Linear	y	x	Product of dependent variable and derivative: $y\frac{dy}{dx}$
(c) $t^5y^{(4)} - t^3y'' + 6y = 0$	ODE	4	Linear	y	t	N/A (y and derivatives are linear; powers are on t)
(d) $\frac{\partial^2u}{\partial x^2} + \frac{\partial^2u}{\partial y^2} = 0$	PDE	2	Linear	u	x, y	N/A (Laplace equation)
(e) $\frac{dy}{dx} = \sqrt{1 + y^2}$	ODE	1	Non-Linear	y	x	Non-linear radical function of y : $\sqrt{1 + y^2}$

 Underlying Material & Intuitive Explanation

[!NOTE] **The 3 Golden Rules of Linearity:** A differential equation in $y(x)$ is **linear** if and only if it can be written as:

$$a_n(x)\frac{d^ny}{dx^n} + a_{n-1}(x)\frac{d^{n-1}y}{dx^{n-1}} + \cdots + a_1(x)\frac{dy}{dx} + a_0(x)y = g(x)$$

1. y and every derivative $y', y'', \dots$ appear strictly to the **first power** (y^1).
2. There are **no products** of y with itself or its derivatives (e.g. $yy', (y')^2, yy''$).
3. y and its derivatives are **not enclosed in transcendental functions** (e.g. $\sin(y), e^y, \ln(y), \sqrt{y}, \frac{1}{y}$).

Crucial: Powers or functions of the independent variable x (like $x^5, \cos x, e^x$) have **zero impact** on linearity! Equation (c) has t^5 , but it is 100% linear because t is independent.

Problem 12 · First-Order Linear ODE via Integrating Factor

Problem Statement

(Dr. Haghighat Lecture 3 Slide 12 & Zill Section 2.3)

Find the unique particular solution of the initial value problem:

$$x \frac{dy}{dx} + 2y = 4x^2, \quad y(1) = 2$$

- (a) Put the equation in standard form and identify $P(x)$ and $Q(x)$.
 - (b) Compute the integrating factor $\mu(x)$.
 - (c) Derive the general solution $y(x)$.
 - (d) Find the particular solution and state the maximal interval of definition I .
-

Step-by-Step Solution

Step 1: Put in Standard Canonical Form

The standard form for a first-order linear ODE is:

$$\frac{dy}{dx} + P(x)y = Q(x)$$

Divide through by the leading coefficient x (for $x \neq 0$):

$$\frac{dy}{dx} + \frac{2}{x}y = 4x$$

Here:

$$P(x) = \frac{2}{x}, \quad Q(x) = 4x$$

Step 2: Compute the Integrating Factor $\mu(x)$

$$\mu(x) = e^{\int P(x)dx} = e^{\int \frac{2}{x}dx} = e^{2 \ln |x|} = e^{\ln(|x|^2)} = e^{\ln(x^2)} = x^2$$

Step 3: Multiply the Standard Form by $\mu(x)$

$$x^2 \left(\frac{dy}{dx} + \frac{2}{x}y \right) = x^2(4x)$$

$$x^2 \frac{dy}{dx} + 2xy = 4x^3$$

Notice that the left-hand side is the exact product rule expansion of $\frac{d}{dx}[\mu(x)y]$:

$$\frac{d}{dx} [x^2 y] = 4x^3$$

Step 4: Integrate Both Sides

$$\int \frac{d}{dx} [x^2 y] dx = \int 4x^3 dx$$

$$x^2 y = x^4 + C$$

Divide by x^2 :

$$y(x) = x^2 + \frac{C}{x^2}$$

Step 5: Apply the Initial Condition $y(1) = 2$

Substitute $x = 1, y = 2$:

$$2 = (1)^2 + \frac{C}{(1)^2} = 1 + C \implies C = 2 - 1 = 1$$

Particular solution:

$$y(x) = x^2 + \frac{1}{x^2} = x^2 + x^{-2}$$

Step 6: Interval of Definition I

The coefficient $P(x) = \frac{2}{x}$ has a discontinuity at $x = 0$. The initial condition is given at $x_0 = 1 > 0$. By the Existence and Uniqueness Theorem for linear equations, the solution exists and is unique on the largest interval containing $x_0 = 1$ on which both $P(x)$ and $Q(x)$ are continuous.

$$I = (0, \infty)$$

💡 Underlying Material & Intuitive Explanation**[!NOTE] Theory & Concepts Tested:**

- Why the Integrating Factor Works:** The integrating factor $\mu(x)$ converts the sum $\mu y' + \mu P y$ into the derivative of a single product $(\mu y)'$ by forcing $\mu' = P\mu \implies \frac{\mu'}{\mu} = P \implies \ln \mu = \int P dx$.
- Standard Form Rule:** NEVER calculate $\mu(x) = e^{\int 2 dx} = e^{2x}$. You **must** divide by the leading coefficient first! Failing to divide by x is the single most common error in Section 2.3.
- Logarithm Exponent Law:** $e^{a \ln x} = e^{\ln(x^a)} = x^a$. A common mistake is writing $e^{2 \ln x} = 2e^{\ln x} = 2x$. The coefficient 2 must move inside the log as a power (x^2).

Part IV: Lecture 4 Exact Equations & Integrating Factors (Quiz Cutoff Topic)

Problem 13 · Exact Differential Equation & Initial Value Problem**Problem Statement**

(Dr. Haghighat Lecture 4 · Slide 9 & Zill Section 2.4)

Consider the first-order initial value problem:

$$\frac{dy}{dx} = \frac{xy^2 - \cos x \sin x}{y(1 - x^2)}, \quad y(0) = 2$$

- (a) Rewrite the differential equation in standard differential form $M(x, y), dx + N(x, y), dy = 0$ and prove that it satisfies the criterion for an exact differential.
- (b) Derive the implicit one-parameter family of solutions $\psi(x, y) = C$ by integrating with respect to x and determining the function of integration $g(y)$.
- (c) Apply the initial condition $y(0) = 2$ to find the constant C .
- (d) Solve explicitly for $y(x)$, justifying your choice of branch ($\pm$), and state the valid interval of definition I containing $x_0 = 0$.

Step-by-Step Solution

Step 1: Put in Differential Form $M(x, y), dx + N(x, y), dy = 0$

Cross-multiply the differential equation:

$$y(1 - x^2), dy = (xy^2 - \cos x \sin x), dx$$

Rearrange all terms to the left-hand side:

$$(\cos x \sin x - xy^2), dx + y(1 - x^2), dy = 0$$

Identify $M(x, y)$ and $N(x, y)$:

$$M(x, y) = \cos x \sin x - xy^2$$

$$N(x, y) = y(1 - x^2) = y - x^2y$$

Step 2: Test Criterion for Exactness

Compute the partial derivative of M with respect to y (treating x as constant):

$$\frac{\partial M}{\partial y} = \frac{\partial}{\partial y} [\cos x \sin x - xy^2] = 0 - 2xy = -2xy$$

Compute the partial derivative of N with respect to x (treating y as constant):

$$\frac{\partial N}{\partial x} = \frac{\partial}{\partial x} [y - x^2y] = 0 - 2xy = -2xy$$

Since $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} = -2xy$ throughout the domain, the differential form is **strictly exact**.

Step 3: Integrate $\frac{\partial \psi}{\partial x} = M(x, y)$ to Find Potential Function $\psi(x, y)$

$$\psi(x, y) = \int M(x, y), dx + g(y) = \int (\cos x \sin x - xy^2), dx + g(y)$$

Evaluate each integral with respect to x :

- For $\int \cos x \sin x, dx$: let $u = \sin x, du = \cos x, dx \implies \int u, du = \frac{1}{2}u^2 = \frac{1}{2} \sin^2 x$.
- For $\int (-xy^2), dx$: treating y as constant, $-y^2 \int x, dx = -\frac{1}{2}x^2y^2$.

$$\psi(x, y) = \frac{1}{2} \sin^2 x - \frac{1}{2} x^2 y^2 + g(y)$$

Step 4: Differentiate with Respect to y and Equate to $N(x, y)$

$$\frac{\partial \psi}{\partial y} = \frac{\partial}{\partial y} \left[\frac{1}{2} \sin^2 x - \frac{1}{2} x^2 y^2 + g(y) \right] = 0 - x^2 y + g'(y)$$

Equate to $N(x, y) = y - x^2y$:

$$-x^2y + g'(y) = y - x^2y$$

Subtract $-x^2y$ from both sides:

$$g'(y) = y$$

Integrate with respect to y :

$$g(y) = \int y, dy = \frac{1}{2}y^2$$

Step 5: State the General Implicit Solution

Substitute $g(y)$ back into $\psi(x, y)$:

$$\psi(x, y) = \frac{1}{2} \sin^2 x - \frac{1}{2}x^2y^2 + \frac{1}{2}y^2 = C_1$$

Multiply through by 2 to clear fractions:

$$\sin^2 x - x^2y^2 + y^2 = C$$

Factor y^2 from the second and third terms:

$$y^2(1 - x^2) + \sin^2 x = C$$

Step 6: Apply Initial Condition $y(0) = 2$

Substitute $x = 0$ and $y = 2$:

$$(2)^2(1 - 0^2) + \sin^2(0) = C$$

$$4(1) + 0 = C \implies C = 4$$

The particular implicit solution is:

$$y^2(1 - x^2) + \sin^2 x = 4$$

Step 7: Isolate Explicit Solution $y(x)$ and Interval of Definition

Isolate y^2 :

$$y^2(1 - x^2) = 4 - \sin^2 x$$

$$y^2 = \frac{4 - \sin^2 x}{1 - x^2}$$

Take square roots:

$$y(x) = \pm \sqrt{\frac{4 - \sin^2 x}{1 - x^2}}$$

Since the initial condition requires $y(0) = +2 > 0$, we select the **positive square root branch**:

$$y(x) = \sqrt{\frac{4 - \sin^2 x}{1 - x^2}}$$

Interval of Definition I : The denominator vanishes when $1 - x^2 = 0 \implies x = \pm 1$. The initial condition is specified at $x_0 = 0 \in (-1, 1)$. On the interval $(-1, 1)$, the denominator $1 - x^2 > 0$. Furthermore, since $0 \leq \sin^2 x \leq 1$, the numerator satisfies $4 - \sin^2 x \geq 3 > 0$, ensuring the radicand is strictly positive. Therefore, the maximal continuous interval of definition

is:

$$\mathbf{I} = (-1, 1)$$

💡 Underlying Material & Intuitive Explanation

[!NOTE] Theory & Concepts Tested:

1. **Potential Function Concept:** An exact equation represents the level curves $\psi(x, y) = C$ of a 3D surface. The total differential is $d\psi = \frac{\partial\psi}{\partial x}dx + \frac{\partial\psi}{\partial y}dy = 0$, which implies $\psi(x, y) = C$.
2. **Symmetry of Mixed Partial (Clairaut's Theorem):** $\frac{\partial M}{\partial y} = \frac{\partial^2\psi}{\partial y\partial x}$ and $\frac{\partial N}{\partial x} = \frac{\partial^2\psi}{\partial x\partial y}$. Continuity guarantees that mixed partials are equal, which is why $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$ is both necessary and sufficient.
3. **The $g(y)$ Disappearance Verification:** When you equate $\frac{\partial\psi}{\partial y} = N(x, y)$, any terms containing x MUST cancel out completely! If you are left with an x inside $g'(y)$, you either made an algebra error when testing exactness or made an integration error in Step 3.

Problem 14 · Non-Exact ODE Made Exact via Integrating Factor

Problem Statement

(Dr. Haghighat Lecture 4 · Slides 10–11 & Zill Section 2.4)

Consider the differential equation:

$$xy, dx + (2x^2 + 3y^2 - 20), dy = 0$$

- (a) Show by calculating $\frac{\partial M}{\partial y}$ and $\frac{\partial N}{\partial x}$ that the equation is **not exact**.
 (b) Evaluate both potential single-variable integrating factor criteria:

$$\frac{M_y - N_x}{N} \quad \text{and} \quad \frac{N_x - M_y}{M}$$

State which expression depends solely on a single variable, and derive the corresponding integrating factor μ .

- (c) Multiply the entire differential equation by μ , verify that the resulting equation is strictly exact, and find the general implicit solution $\psi(x, y) = C$.

Step-by-Step Solution

Step 1: Identify M , N and Test for Exactness

$$M(x, y) = xy$$

$$N(x, y) = 2x^2 + 3y^2 - 20$$

Compute partial derivatives:

$$\frac{\partial M}{\partial y} = x$$

$$\frac{\partial N}{\partial x} = 4x$$

Since $\frac{\partial M}{\partial y} = x \neq 4x = \frac{\partial N}{\partial x}$, the equation is **not exact**.

Step 2: Test Integrating Factor Formulas**Test 1 (Function of x alone?):**

$$\frac{M_y - N_x}{N} = \frac{x - 4x}{2x^2 + 3y^2 - 20} = \frac{-3x}{2x^2 + 3y^2 - 20}$$

This expression depends on both x and y . No single-variable integrating factor $\mu(x)$ exists.

Test 2 (Function of y alone?):

$$\frac{N_x - M_y}{M} = \frac{4x - x}{xy} = \frac{3x}{xy} = \frac{3}{y}$$

This expression depends **solely on y** !

Step 3: Compute the Integrating Factor $\mu(y)$

$$\mu(y) = e^{\int \left(\frac{N_x - M_y}{M}\right) dy} = e^{\int \frac{3}{y} dy} = e^{3 \ln |y|} = e^{\ln(|y|^3)} = y^3$$

(Note: Choosing the positive branch $\mu(y) = y^3$ suffices to make the equation exact).

Step 4: Multiply the Original Equation by $\mu(y) = y^3$

$$y^3[xy, dx + (2x^2 + 3y^2 - 20), dy] = 0$$

$$(xy^4), dx + (2x^2y^3 + 3y^5 - 20y^3), dy = 0$$

Now define the new coefficient functions:

$$M_{new}(x, y) = xy^4$$

$$N_{new}(x, y) = 2x^2y^3 + 3y^5 - 20y^3$$

Verify exactness of the new equation:

$$\frac{\partial M_{new}}{\partial y} = \frac{\partial}{\partial y}[xy^4] = 4xy^3$$

$$\frac{\partial N_{new}}{\partial x} = \frac{\partial}{\partial x}[2x^2y^3 + 3y^5 - 20y^3] = 4xy^3$$

Since $\frac{\partial M_{new}}{\partial y} = \frac{\partial N_{new}}{\partial x} = 4xy^3$, the transformed differential equation is **strictly exact**!

Step 5: Integrate M_{new} with respect to x to find $\psi(x, y)$

$$\psi(x, y) = \int M_{new}, dx + g(y) = \int (xy^4), dx + g(y) = \frac{1}{2}x^2y^4 + g(y)$$

Step 6: Differentiate with respect to y and equate to N_{new}

$$\frac{\partial \psi}{\partial y} = \frac{\partial}{\partial y} \left[\frac{1}{2}x^2y^4 + g(y) \right] = 2x^2y^3 + g'(y)$$

Equate to N_{new} :

$$2x^2y^3 + g'(y) = 2x^2y^3 + 3y^5 - 20y^3$$

Cancel the common term $2x^2y^3$:

$$g'(y) = 3y^5 - 20y^3$$

Step 7: Integrate $g'(y)$ with respect to y

$$g(y) = \int (3y^5 - 20y^3), dy = 3 \left(\frac{y^6}{6} \right) - 20 \left(\frac{y^4}{4} \right) = \frac{1}{2}y^6 - 5y^4$$

Step 8: Assemble the General Solution $\psi(x, y) = C$

Substitute $g(y)$ back into $\psi(x, y)$:

$$\psi(x, y) = \frac{1}{2}x^2y^4 + \frac{1}{2}y^6 - 5y^4 = C_1$$

Multiply through by 2 to clear fractions:

$$x^2y^4 + y^6 - 10y^4 = C$$

Factoring out y^4 :

$$y^4(x^2 + y^2 - 10) = C$$

Underlying Material & Intuitive Explanation

[!NOTE] **Theory & Concepts Tested:**

1. The Formula Sign Trap:

- When dividing by N , the numerator is $(M_y - N_x)$, leading to $\mu(x) = e^{\int \frac{M_y - N_x}{N} dx}$.
- When dividing by M , the numerator is REVERSED: $(N_x - M_y)$, leading to $\mu(y) = e^{\int \frac{N_x - M_y}{M} dy}$.
- *Mnemonic:* The subtracted partial corresponds to the denominator:
 - Denominator $N \implies$ subtract N_x .
 - Denominator $M \implies$ subtract M_y .

2. **Always Re-Verify Exactness:** After multiplying by μ , compute $\frac{\partial M_{new}}{\partial y}$ and $\frac{\partial N_{new}}{\partial x}$ before integrating. It takes only 10 seconds and guarantees that your integrating factor was 100% correct.